

Mutation

1 Создаем таблицу

```
CREATE DATABASE IF NOT EXISTS task_11;

CREATE TABLE IF NOT EXISTS task_11.user_activity
(
    user_id      UInt32,
    activity_type String,
    activity_date DateTime
)
ENGINE = MergeTree
    ORDER BY (user_id)
    PARTITION BY toYYYYMM(activity_date);
```

```
CREATE DATABASE IF NOT EXISTS task_11
```

Query id: 8abd332d-f154-4d96-a859-a0c41a2f79ea

Ok.

0 rows in set. Elapsed: 0.019 sec.

```
CREATE TABLE IF NOT EXISTS task_11.user_activity
(
    `user_id`      UInt32,
    `activity_type` String,
    `activity_date` DateTime
)
ENGINE = MergeTree
PARTITION BY toYYYYMM(activity_date)
ORDER BY user_id
```

Query id: 2dc1079f-74e2-46c5-88db-b7e13be82a2f

Ok.

0 rows in set. Elapsed: 0.019 sec.

2. Наполняем ее данными

```
INSERT INTO task_11.user_activity
SELECT toUInt32(rand() % 10000 + 1) AS user_id,
       ['login', 'logout', 'purchase'][(rand() % 3) + 1] AS activity_type,
       (date + toIntervalSecond(rand() % 86400)) AS activity_date
FROM (SELECT today() - number AS date
      FROM numbers(180)) AS dates
      CROSS JOIN (SELECT *
                  FROM numbers(50)) AS nums;
```

```
INSERT INTO task_11.user_activity SELECT
    toUInt32((rand() % 10000) + 1) AS user_id,
    ['login', 'logout', 'purchase'][(rand() % 3) + 1] AS activity_type,
    date + toIntervalSecond(rand() % 86400) AS activity_date
FROM
(
    SELECT today() - number AS date
    FROM numbers(180)
```

```

) AS dates
CROSS JOIN
(
    SELECT *
    FROM numbers(50)
) AS nums

```

Query id: 9013e4f1-965d-4bb4-828d-3d5327afccae

Ok.

0 rows in set. Elapsed: 0.013 sec.

3. Мутируем данные

```

SELECT *
FROM task_11.user_activity
WHERE user_id IN (500, 1004)
AND toDate(activity_date) < '2025-11-30';

```

```

SELECT *
FROM task_11.user_activity
WHERE (user_id IN (500, 1004)) AND (toDate(activity_date) < '2025-11-30')

```

Query id: d557bef4-727b-4234-b320-f4a9a6ab526a

↑ Progress: 8.85 thousand rows, 118.18 KB (739.56 thousand rows/s., 9.88 MB/s.) ↗ Progress: 8.85 thousand rows, 118.18 KB (739.56 thousand rows/s., 9.88 MB/s.)

	user_id	activity_type	activity_date
1.	500	login	2025-08-31 07:41:39
2.	1004	purchase	2025-10-15 20:23:23

2 rows in set. Elapsed: 0.012 sec. Processed 8.85 thousand rows, 118.18 KB (728.19 thousand rows/s., 9.72 MB/s.)

Peak memory usage: 124.17 KiB.

```

ALTER TABLE task_11.user_activity
UPDATE activity_type = 'login'
WHERE user_id IN (500, 1004)
AND toDate(activity_date) < '2025-11-30';

```

```

ALTER TABLE task_11.user_activity
(UPDATE activity_type = 'login' WHERE (user_id IN (500, 1004)) AND (toDate(activity_date) < '2025-11-30'))

```

Query id: 25f2c032-711a-454c-9fd4-4f569ff9998f

Ok.

0 rows in set. Elapsed: 0.013 sec.

Выполняем выборку данных

```

SELECT *
FROM task_11.user_activity
WHERE user_id IN (500, 1004)
AND toDate(activity_date) < '2025-11-30';

```

```

SELECT *
FROM task_11.user_activity
WHERE (user_id IN (500, 1004)) AND (toDate(activity_date) < '2025-11-30')

```

Query id: 5db26ba7-20d8-4254-b90c-36273e765e6a

↓ Progress: 8.85 thousand rows, 118.18 KB (1.08 million rows/s., 14.36 MB/s.) 9✓ Progress: 8.85 thousand rows, 118.18 KB (1.08 million rows/s., 14.36 MB/s.) 9

	user_id	activity_type	activity_date
1.	1004	login	2025-10-15 20:23:23
2.	500	login	2025-08-31 07:41:39

2 rows in set. Elapsed: 0.008 sec. Processed 8.85 thousand rows, 118.18 KB (1.06 million rows/s., 14.16 MB/s.)
Peak memory usage: 212.99 KiB.

```
SELECT *  
  FROM system.mutations  
FORMAT Vertical;
```

```
SELECT *  
FROM system.mutations  
FORMAT Vertical
```

Query id: 74d74404-f4a5-4432-8db2-3b0dcd4bb02a

Row 1:

database:	default
table:	github_events_enriched
mutation_id:	mutation_53.txt
command:	(MATERIALIZED PROJECTION proj_distinct_actor_login)
create_time:	2025-12-02 18:39:13
block_numbers.partition_id:	['']
block_numbers.number:	[53]
parts_to_do_names:	[]
parts_to_do:	0
is_done:	1
is_killed:	0
latest_failed_part:	
latest_fail_time:	1970-01-01 00:00:00
latest_fail_reason:	
latest_fail_error_code_name:	

Row 2:

database:	task_11
table:	user_activity
mutation_id:	mutation_8.txt
command:	(UPDATE activity_type = 'login' WHERE (user_id IN (500, 1004)) AND (toDate(activity_date) < '2025-11-30'))
create_time:	2025-12-03 21:37:43
block_numbers.partition_id:	['']
block_numbers.number:	[8]
parts_to_do_names:	[]
parts_to_do:	0
is_done:	1
is_killed:	0
latest_failed_part:	
latest_fail_time:	1970-01-01 00:00:00
latest_fail_reason:	
latest_fail_error_code_name:	

2 rows in set. Elapsed: 0.003 sec.

4 Манипуляции с партициями

```
SELECT database,  
       table,  
       partition,  
       rows,  
       active  
  FROM system.parts  
 WHERE table = 'user_activity'  
 ORDER BY partition DESC;
```

```

SELECT
    database,
    `table`,
    partition,
    rows,
    active
FROM system.parts
WHERE `table` = 'user_activity'
ORDER BY partition DESC

```

Query id: 35a32611-0d9a-44b0-99ef-73eecb39cdf9

	database	table	partition	rows	active
1.	task_11	user_activity	202512	150	0
2.	task_11	user_activity	202512	150	1
3.	task_11	user_activity	202511	1500	0
4.	task_11	user_activity	202511	1500	1
5.	task_11	user_activity	202510	1550	0
6.	task_11	user_activity	202510	1550	1
7.	task_11	user_activity	202509	1500	0
8.	task_11	user_activity	202509	1500	1
9.	task_11	user_activity	202508	1550	0
10.	task_11	user_activity	202508	1550	1
11.	task_11	user_activity	202507	1550	0
12.	task_11	user_activity	202507	1550	1
13.	task_11	user_activity	202506	1200	0
14.	task_11	user_activity	202506	1200	1

14 rows in set. Elapsed: 0.003 sec.

```

ALTER TABLE task_11.user_activity DROP PARTITION '202509';

```

```

ALTER TABLE task_11.user_activity
(DROP PARTITION '202509')

```

Query id: 1a4dce5b-0972-4ee9-950f-f75105f29280

Ok.

0 rows in set. Elapsed: 0.003 sec.

```

SELECT database,
    table,
    partition,
    rows,
    active
FROM system.parts
WHERE table = 'user_activity'
ORDER BY partition DESC;

```

```

SELECT
    database,
    `table`,
    partition,
    rows,
    active
FROM system.parts
WHERE `table` = 'user_activity'
ORDER BY partition DESC

```

Query id: a641cf7f-bd50-4f30-b1fd-da96c8ef2052

	database	table	partition	rows	active
1.	task_11	user_activity	202512	150	0
2.	task_11	user_activity	202512	150	1
3.	task_11	user_activity	202511	1500	0

4.	task_11	user_activity	202511	1500	1
5.	task_11	user_activity	202510	1550	0
6.	task_11	user_activity	202510	1550	1
7.	task_11	user_activity	202509	1500	0
8.	task_11	user_activity	202509	0	0
9.	task_11	user_activity	202508	1550	0
10.	task_11	user_activity	202508	1550	1
11.	task_11	user_activity	202507	1550	0
12.	task_11	user_activity	202507	1550	1
13.	task_11	user_activity	202506	1200	0
14.	task_11	user_activity	202506	1200	1

14 rows in set. Elapsed: 0.004 sec.

Заголовок